



# INFORMATIQUE

## CHAPTER 1 - INTRODUCTION

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Faculty of Medicine – University of Setif 1



## Objective and plan

### ► Objective :

- Prerequisites and upgrading

### ► Plan :

- Introduction
- Material
- Operating system
- Computer languages
- Applications
- Data representation
- Data size measurement units

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## Introduction to Computer Science

- **Definition:** Techniques for the automatic processing of **information** using computers

### ► Information processing

- Calculate
- Transform
- Dematerialize
- Store
- Transfer

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## Introduction to Computer Science

### Elements of a Computer System

#### Hardware

(PC, Mac, Tablet, Workstation, etc...)

#### Operating system

(Windows , Mac OS, Android , Linux, etc...)

#### Languages

(Java, C/C++, Fortran, Python, etc.)

#### Applications

(Word, Excel, Games, etc...)

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## Hardware: Main Components of a PC

### ► Central unit (the case)

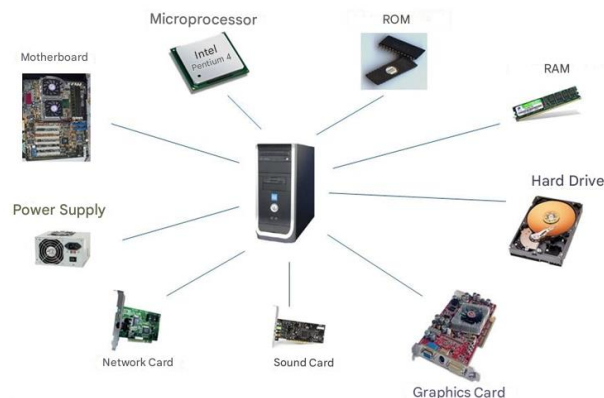
- Processor or CPU (Central Processing Unit)
- Main memory (RAM)
- Storage media (HDD, SSD, CD/DVD, flash drive, etc.)
- Specialized cards (video cards, network cards, etc.)
- Input/Output Interfaces (Ports)

### ► Peripherals

- Input: keyboard, mouse, microphone,...
- Output: screen, printer, speakers,...
- Input/Output: modem, headphones,...

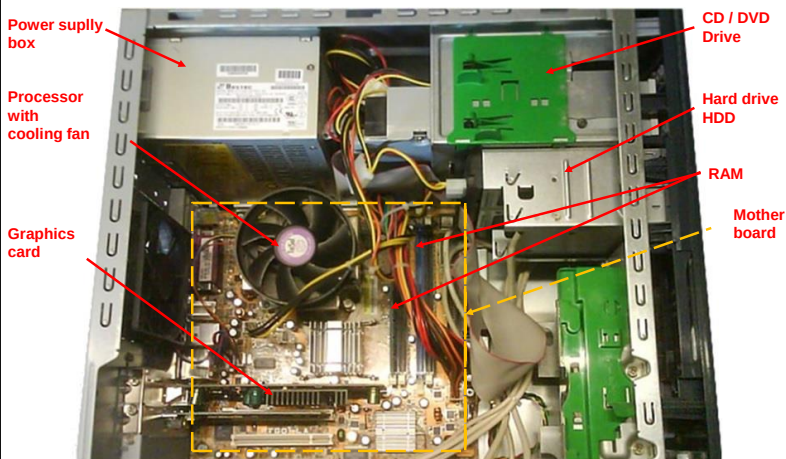
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## Hardware: The Central Unit



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## Hardware: The Central Processing Unit



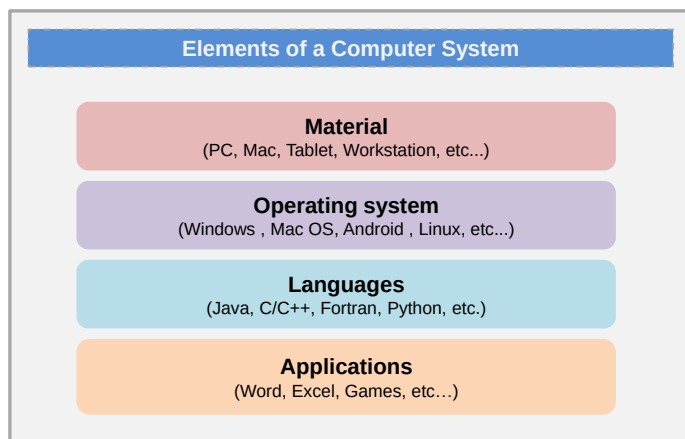
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## Hardware: Ports (peripheral connections)

| Connectors | Descriptions                                                          |
|------------|-----------------------------------------------------------------------|
|            | Mouse                                                                 |
|            | Keyboard                                                              |
|            | USB port for devices with USB connectors                              |
|            | Printer                                                               |
|            | Monitor                                                               |
|            | Audio input and output lines                                          |
|            | Microphone                                                            |
|            | Headphones                                                            |
|            | FireWire (IEEE 1394) for cameras or other very high-speed peripherals |

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## Introduction to Computer Science



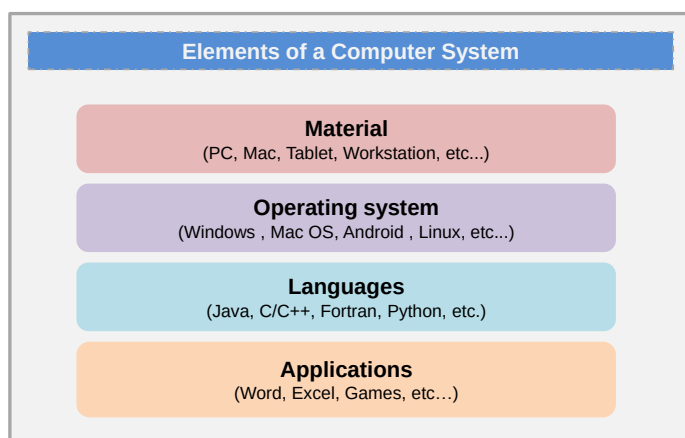
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## What Is An Operating System?

- A Operating System is a set of programs that manage hardware and control applications
  - Device management (screen display, keyboard reading, printer control, etc.)
  - User and data management (accounts, resource sharing, file and directory management, etc.)
  - User interface (textual or graphical): Interpretation of commands
  - Program control (task splitting, CPU time sharing, etc.)

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## Introduction to Computer Science



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## Computer languages

- A computer language is a tool used to give orders (**instructions**) to the machine.
  - Each instruction corresponds to a processor action
- Purpose: write programs (a consecutive sequence of instructions) designed to perform a given task
  - Example: a pharmaceutical product management program
- Constraint: must be understandable by the machine

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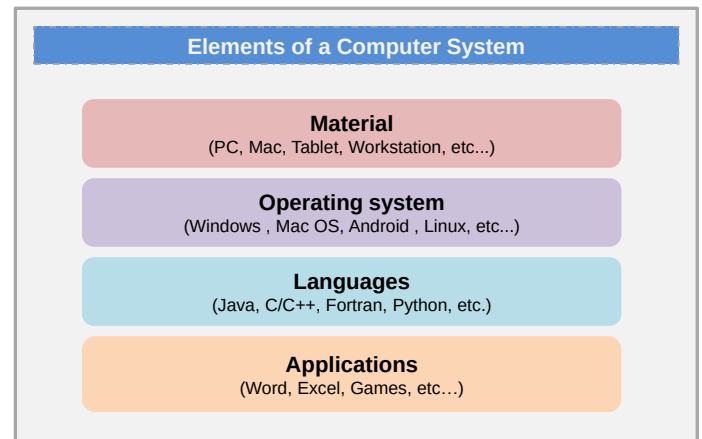
## High Level Computer Languages

- ▶ **Multiple benefits for the High Level:**
  - ▶ Close to human language "English" (understandable)
  - ▶ Allows for greater portability (independent from hardware)
  - ▶ Manipulation of complex data and expressions (reals, objects,  $a*b/c$ , ...)
- ▶ **Need for a translator (Compiler/Interpreter)**



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## Introduction to Computer Science



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## User Applications

- ▶ **Office applications:**
  - ▶ Word (Writing)
  - ▶ Excel (Calculation)
  - ▶ PowerPoint (Presentation)
- ▶ **Communication and internet applications :**
  - ▶ Messaging
  - ▶ Social networks
  - ▶ Web browsers

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## Medical Applications (1)

- ▶ **Medical database management**
  - ▶ Patients, Medications, Diseases
- ▶ **Medical facility management**
  - ▶ Hospitals, Pharmacies, Labs, Doctors' offices
- ▶ **Control, monitoring and prevention**
  - ▶ Epidemic Surveillance Center
  - ▶ Pharmacovigilance

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## Medical Applications (2)

- ▶ **Artificial intelligence**
  - ▶ Medical diagnosis
- ▶ **Virtual medical simulation**
  - ▶ Training and learning in surgery
- ▶ **Medical imaging**
  - ▶ X-rays, Ultrasounds, MRI, and CT Scans
- ▶ **Telemedicine**
  - ▶ Remote monitoring, teleconsultation, telesurgery

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## Objective and plan

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  - ▶ Applications
  - ▶ Date representation
  - ▶ Data size measurement units

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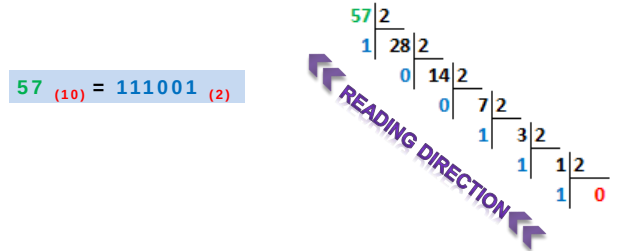
## Data representation (1)

- ▶ On computers, data is represented and saved on physical media (memory) in binary.
- ▶ This is not a random choice, but a current technological limitation.
- ▶ Constraint: everything must be represented using the symbols 0 and 1
  - ▶ Text
  - ▶ Picture
  - ▶ Sound
  - ▶ ... Etc.

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## Data representation (2)

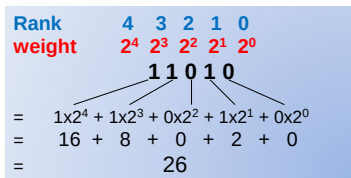
- ▶ Converting decimal to binary
  - ▶ Perform successive integer divisions by 2 until the quotient becomes zero.
- ▶ Conversion example:
  - ▶ 57 (decimal base) = 111001 (binary base)



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## Data representation (3)

- ▶ Converting binary to decimal
  - ▶ To convert a binary number to a decimal number, you must multiply the value of each bit by its weight, then add the results together.
- ▶ Conversion example:
  - ▶ 11010 (2) =  $1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 26$  (10)



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## Data size measurement units

- ▶ The primary unit of measurement for memory size is the byte (8 bits).
- ▶ A bit is the elementary binary unit (0 or 1)

| Unit name | Designation | Value and equivalent                                                                               |
|-----------|-------------|----------------------------------------------------------------------------------------------------|
| Bit       | b           | $2^0 \text{ b} = 1 \text{ bit}$ (Elementary unit)                                                  |
| Byte      | B           | $2^3 \text{ b} = 8 \text{ bits} = 1 \text{ Byte}$ (Primary Unit)                                   |
| Kilobyte  | KB          | $2^{10} = 1024 \text{ bytes}$                                                                      |
| Megabyte  | MB          | $2^{20} \text{ B} = 2^{10} \text{ KB}$                                                             |
| Gigabyte  | GB          | $2^{30} \text{ B} = 2^{10} \text{ MB} = 2^{20} \text{ KB}$                                         |
| Terabyte  | TB          | $2^{40} \text{ B} = 2^{10} \text{ GB} = 2^{20} \text{ MB} = 2^{30} \text{ KB}$                     |
| Petabyte  | PB          | $2^{50} \text{ B} = 2^{10} \text{ TB} = 2^{20} \text{ GB} = 2^{30} \text{ MB} = 2^{40} \text{ KB}$ |

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